

題號 4 (10分)

(a) $\lim_{x \rightarrow 0} \frac{x^2 - 4}{x^2 - 2x + 2} = \frac{0}{2} = 0$ (1 point)

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{x+1}}}{1} = \frac{1}{2}$ (1 point)

(c) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(d) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(e) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(f) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(g) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(h) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(i) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

(j) $\lim_{x \rightarrow 0} \frac{x^2 - 1}{x^2 + 1} = \frac{-1}{1} = -1$ (1 point)

題號 1 (6分)

(a) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

(b) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

(c) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

(d) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

(e) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

(f) $\lim_{x \rightarrow 0} \frac{\ln|x+1|}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{x+1}}{1} = 1$ (1 point)

題號 1 (4分)

(a) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(b) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(c) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(d) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

題號 1 (10分)

(a) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(b) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(c) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(d) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(e) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(f) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(g) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(h) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(i) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

(j) $\lim_{x \rightarrow 0} \frac{x^2 + 2x + 9}{x^2 + 5x + 1} = \frac{9}{1} = 9$ (1 point)

題號 2 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 2 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 2 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 2 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 3 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 3 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 3 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 3 (10分)

(a) $f(x) = \frac{1}{x-1} + x$

(b) Find the horizontal asymptote of $f(x)$. (1 point)

(c) Find the vertical asymptote of $f(x)$. (1 point)

(d) Find the horizontal asymptote of $f(x)$. (1 point)

(e) Find the vertical asymptote of $f(x)$. (1 point)

(f) Find the horizontal asymptote of $f(x)$. (1 point)

(g) Find the vertical asymptote of $f(x)$. (1 point)

(h) Find the horizontal asymptote of $f(x)$. (1 point)

(i) Find the vertical asymptote of $f(x)$. (1 point)

(j) Find the horizontal asymptote of $f(x)$. (1 point)

題號 6 (6分)

(a) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(b) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(c) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(d) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(e) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(f) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

題號 6 (6分)

(a) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(b) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(c) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(d) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(e) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

(f) Find the "open" interval on which the function $f(x) = (x^2 + 2)e^x$ is increasing and those on which is decreasing. (6 points)

國立台灣師範大學 113 年度上學期微積分乙(一)期中考

考試須知:

1. 本試卷共六題(共二頁), 配分請見各題, 合計 100 分。
2. 禁止攜帶書表、筆記和資料等進考場。
3. 禁止使用手機、計算機及電腦等電子用品計算題目和獲得答案。
4. 所有题目的解答都要有計算過程, 無計算過程不給分。

- 1.(10分) Find the limits (Write or ∞ where appropriate). Note that L'Hopital's Rule is forbidden.
- 計算以下函數的極限, 必要時請寫成 ∞ 不可使用羅必達

(a) $\lim_{x \rightarrow 0} \frac{x-4}{x^2-5x+4}$ (b) $\lim_{x \rightarrow 0} \frac{\sqrt{x}-\sqrt{x^2+4}}{x}$ (c) $\lim_{x \rightarrow 0} \frac{\ln 2x}{5x}$

(d) $\lim_{x \rightarrow 0} \frac{3x^2+5x+9}{5x^3+x+1}$ (e) $\lim_{x \rightarrow 1} \frac{x^3-1}{x+1}$ (f) $\lim_{x \rightarrow 2} \frac{3x}{2x-10}$

- 2.(10分) Find the horizontal and vertical asymptote of the function.
- 求以下函數的水平漸近線和垂直漸近線

$f(x) = \frac{1}{x^2-9} + 3$

- 3.(30分) Find the derivatives of the following functions dy/dx .
- 計算以下函數的導數

(a) $y = \frac{x^2}{4x-3}$ (b) $y = \sqrt{2x^3-5x}$ (c) $y = (\sec x)^3$

(d) $y = \arctan|x|, x \neq 0$ (e) $y = \sqrt{\frac{x-1}{x-2}}$ (f) $y = \log_2 x$

- 4.(10分) Use implicit differentiation to find dy/dx .
- 使用隱函數微分法求 dy/dx

$x^2 + y^2 = \sin(xy)$

- 5.(10分) Show that the equation $x^2 + 4x + 1 = 0$ has exactly one solution.
- 證明方程式 $x^2 + 4x + 1 = 0$ 只有一個實數解。

- 6.(10分) Find the open intervals on which the function f is increasing and those on which is decreasing.
- 求以下函數遞增的區間和遞減的區間

$f(x) = (x^2 - 2)e^{2x}$